TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

BIOLOGY 1

033/1

(For both School and Private Candidates)

TIME: 3 HOURS **Thursday, 22nd August 2024 a.m.**

Instructions

1. This paper consists of three sections A, B and C with a total of eleven (11) questions.
2. Answers **all** questions in sections A and B and **two (2)** questions from section C.
3. Except diagrams that must be drawn in pencil all writings should be in blue or black ink.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s) provided.

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in answer booklet (s) provided.

   (i) Suppose you have been assigned by your School Head to cross-section cutting down the mango stem and then recommend on the arrangement of vascular bundle. Which one would you recommend?
   A. Random pattern
   B. Scattered pattern
   C. Ring pattern
   D. Peripheral pattern
   E. Zigzag pattern

   (ii) Mr. Araali wants to export glass materials from Tanzania to DRC but he fears on its safety of his luggage. Recommend the biological term that he could put on item's box as a safety sign.
   A. Fragile
   B. Biohazard
   C. Corrosive
   D. Flammable
   E. Toxic

   (iii) A patient has been diagnosed with low level of blood sugar. Which hormone would you recommend to regulate the victim's sugar?
   A. Insulin
   B. Glucagon
   C. Antidiuretic
   D. Aldosterone
   E. Testosterone

   (iv) Mrs. Akena is a pregnant of seven months, she was traveling to Arusha for burial ceremony of her cousin, on the way they got accident and she suddenly started experiencing labour pain and when brought to hospital she delivered a baby. Which term is used to refer the condition that occurred to Mrs. Akena?
   A. Abortion
   B. Still birth
   C. Premature birth
   D. Breech birth
   E. Induced birth

   (v) _____are considered to be very successful group of animals on earth due to their diversity in life style, feeding methods and even movement.
   A. Protoctista
   B. Invertebrates
   C. Mollusks
   D. Insects
   E. Arachnida

(vi) As a botanist what would you advise the student to use in his experiment to observe different stages of mitosis?

A. Leaf
B. Root tip
C. Stem
D. Ovary
E. Flower

(vii) Being a footballer Atwoki is likely seen to lose a lot of water through sweating. As upcoming doctor, identify the organ of the body which is responsible for the loss?

A. Lungs
B. Skin
C. Kidney
D. Heart
E. Liver

(viii) Doctors emphasized women to breastfeed their babies to strengthen their body immunity. The aim of doctor's advice is to help babies to acquire:

A. Natural passive immunity
B. Natural active immunity
C. Artificial passive immunity
D. Artificial active immunity
E. Active passive immunity

(ix) Pa Amadu experienced frequent and burning urination, smelling urine, inability to empty bladder completely and leg cramp. What common disorder of the excretory system would you advise him to diagnose?

A. Uraemia
B. Kidney or renal failure
C. Gout
D. Urinary Tract Infection (UTI)
E. Kidney stones (renal calculi)

(x) In monohybrid crossing diagrams, the homozygous dominant genotype is represented by _____

A. A pair of small letters of the same letter in the alphabet
B. One capital letter and one small letter of the same letter in the alphabet
C. A pair of capital letters of the same letter in the alphabet
D. Two capital letters of different letters in the alphabet
E. Two small letters of different letters in the alphabet

2. Match the process of the mitosis in LIST A with the correct stage of each process in LIST B by writing the letter of the correct response beside the item number.

LIST A		LIST B
(i)	Chromosomes line up at the equator of the spindle fibre	A. Prophase
(ii)	Division of the cytoplasm to form two new cells	B. Telophase
(iii)	Chromatids reach to opposite poles	C. Metaphase
(iv)	A non-dividing stage	D. Cytokinesis
(v)	Chromatids tend to shorten, thicken and become visible	E. Interphase
(vi)	The centromere split and spindle fibres contract, pulling the chromatids to the opposite pole of the cell.	F. Anaphase
		G. Cell cycle
		H. Crossing over

SECTION B (54 Marks)

Answer **all** questions in this section

3. After a heavy rain season, a house of Mr. Zaki was surrounded with stagnant water with green scum.
   (a) Identify the micro organisms that cause green scum on water.
   (b) Draw a well labelled diagram of micro organism identified in 3(a) above and label the parts with the following functions:
      (i) Movement (locomotion)
      (ii) Regulate water content
      (iii) Photosynthesis process

4. (a) In natural environment organisms do interact with one another and with non-living things so as to maintain the ecosystem. Briefly explain three (3) symbiotic ways in which organisms use to interact.

   (b) In a certain ecosystem there were the following organisms: grasses, grasshopper. Toad, Bacteria and Snake. Construct the feeding relationship that will accommodate all organisms.

   (c) Identify trophic levels for each organism in the feeding relationship above.

5. Movement of water in the animal's body depends on concentration gradient.
   (a) Define the natural phenomenon that reflect the above statement.

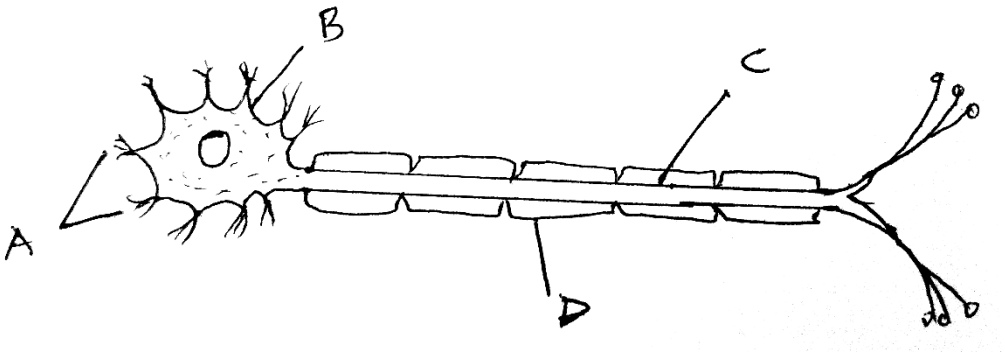
   (b) With the aid of diagrams, explain how the phenomenon in 5(a) above can affect the animal cells.

6. (a) It is not true that the function of human being skeleton is only to determine the shape like most of people say. By using the knowledge of biology you have, support the above statement with five (5) points.

   (b) Highlight two differences between skeletal muscles and Cardiac muscles.

7. Sickle cell anaemia is a hereditary disease due to a recessive gene, which changes normal haemoglobin (HbA) to abnormal haemoglobin (HbS). The red blood cells of people with sickle cell anaemia are sickle shaped.
   (a) What are the possible phenotypes of the offsprings of a man who is heterozygous and a woman who also heterozygous?
   (b) What proportion of the offsprings would be;
      (i) Severely anaemic?
      (ii) Mild anaemia?

8. (a) Study the diagram below on the structure of a nerve cell and answer the questions that follow:



      (i) Name the parts labelled A, B, C and D.
      (ii) Where does the neurone in the diagram carry impulses to?
      (iii) Name two other types of nerve cells found in the nervous system of vertebrates and state their functions.

   (b) State two difference between nervous communication and endocrine communication.

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. Food guide pyramid is not universal for all groups of people, it depends on the needs and health status of the people. By using two examples of food for each case prove this statement with reference of the following:
   (a) Elders
   (b) Pregnancy and expectant mother
   (c) Sedentary workers
   (d) Manual workers
   (e) Sick persons

10. The interaction of biotic and abiotic components of the environment is important for the natural cycles. How would you justify the statement by explaining seven stages involved in circulation of water in the environment.

11. It is just a human nature for the woman to experience monthly discharge of the uterus lining, dead ovum and epithelial cells of the uterus as debris out of the vagina.
    (a) What biological term you would use and explain the biological phenomenon above.
    (b) The process in (a) above is controlled by four hormones and each hormone performs its specific roles. As biology student how would you justify the statement?
    (c) As an expert in food and nutrition department, which types of food would you advise the women to use for the replacement of the blood being discharged during the process in (a) above?

Wabillah Tawfiq